SemEval 2022 Task 2 Subtask A: Multilingual Idiomaticity Detection

Valhari Meshram
University of Potsdam
Potsdam, Germany
meshram@uni-potsdam.de

Abstract

Multiword expressions (MWEs) pose a fundamental challenge for natural language processing because their meaning often cannot be derived from their constituent words. Idiomaticity detection involves determining whether an MWE is used idiomatically or literally in context, requires context-sensitive disambiguation rather than surface-level pattern matching. We study this problem as a binary classification task across English, Portuguese, and Galician under zero-shot and one-shot settings, comparing surface-level logistic regression baselines against fine-tuned multilingual transformers (mBERT and ModernBERT). Model behaviour is analysed across input representations, linguistic feature augmentations, and diagnostic evaluation slices designed to separate genuine contextual reasoning from frequency-driven memorisation. Our results show that surface features, while partially effective in English, fail completely in Portuguese, highlighting the inadequacy of lexical patterns for cross-lingual idiomaticity detection. Contextual models overcome this gap, demonstrating genuine sense disambiguation even on minority-sense instances where surface baselines systematically fail. Cross-lingual transfer improves when languages are typologically related, but remains limited across more distant pairs. Linguistic feature augmentations, including named entity tags and word glosses, do not provide consistent gains, suggesting that contextual representations already encode the relevant information. Overall, our findings confirm that idiomaticity detection is a fundamentally context-dependent problem requiring representations that go beyond surface statistics.

1 Introduction

Multiword expressions (MWEs), particularly idioms, pose a fundamental challenge for natural language processing, as their meaning often cannot be derived directly from their constituent words. Many MWEs are *potentially idiomatic*, meaning the same expression can be interpreted either literally or idiomatically depending on context. For example, a system that treats *eye candy* as inherently idiomatic will fail when encountering *Eye Candy Motorsports* as a named entity. This context dependence makes idiomaticity detection difficult and impacts tasks such as machine translation, question answering, semantic parsing, and information extraction. A key property of idiomaticity is that it exists on a continuum rather than a strict division between compositional and non-compositional expressions (Fazly et al., 2009). As a result, systems must determine meaning from context rather than rely on the expression alone. In zero-shot settings, this requires generalising to unseen MWEs, while in ambiguous cases, models must overcome biases toward the more frequent sense.

Despite the success of pre-trained language models, prior work suggests that their performance may rely on surface patterns and frequency biases rather than robust contextual understanding. It remains unclear to what extent neural language models truly capture idiomatic meaning or primarily rely on memorisation and statistical regularities (Kim et al., 2025).

The SemEval-2022 Task 2 Subtask A (Madabushi et al., 2022) formulates idiomaticity detection as a multilingual classification problem across English, Portuguese, and Galician, under both zero-shot and one-shot conditions. These settings explicitly test generalisation beyond memorised expressions and the effect of minimal supervision. Models that rely on dominant-sense memorisation are expected to fail on the most challenging cases, namely, ambiguous expressions and minority-sense instances.

We study idiomaticity detection by comparing a logistic regression baseline based on lexical features with fine-tuned multilingual transformer models (mBERT and modernBERT), and analyse their behaviour under different input representations and feature augmentations. Our ablations are guided by five research questions: (1) how far surface-level features can carry idiomaticity detection, and where they fail; (2) whether contextual models gen-

uinely disambiguate meaning or rely on frequency biases; (3) how linguistically motivated features affect performance across languages; (4) how much does one-shot learning help; and (5) to what extent idiomaticity knowledge transfers across languages.

2 Related Work

Madabushi et al. (2022) show that transformer-based models, often combined with linguistically motivated features, achieve strong performance across languages. They also highlight differences between zero-shot and one-shot settings, reflecting the distinct challenges of generalising to unseen MWEs and adapting with minimal supervision. Building on these observations, we construct diagnostic evaluation slices that quantify where contextual representations add value over surface patterns, separating ambiguous MWEs from unambiguous ones and analysing model behaviour on each.

Hauer et al. (2022) investigate augmenting inputs with glosses of MWE components, based on the intuition that idiomatic meanings differ from the meanings of individual words. This approach introduces explicit lexical semantic information into the model input. We incorporate this idea as one of the input variants in our experiments to examine how such signals interact with contextual representations.

Another line of work models idiomaticity as a form of semantic divergence between an expression and its context. Tedeschi and Navigli (2022) propose a dual-encoder architecture that measures similarity between MWE and context representations, and show that named entities are a common source of errors. They address this by applying Named Entity Recognition (NER) as a hard filtering step, directly classifying entity-overlapping expressions as literal. We draw on this insight by incorporating NER as a soft input augmentation, allowing the model to use entity signals alongside contextual cues and analyse its effect across languages.

Madabushi et al. (2021) show that pre-trained language models partially capture idiomatic meaning but struggle with genuinely ambiguous expressions. Their findings suggest the need to evaluate model behaviour beyond aggregate metrics. Building on this, we operationalise ambiguity by constructing evaluation slices that separate ambiguous MWEs from unambiguous ones, enabling a more precise analysis of contextual disambiguation and surface-frequency effects.

Building on these insights, we focus on analysing idiomaticity detection through controlled evaluation settings and input variations, with particular emphasis on understanding how contextual representations behave under ambiguity, variation in supervision, and multilingual conditions.

3 Task Formalisation

Idiomaticity detection is formulated as a binary classification problem over multiword expressions (MWEs) in context, across English, Portuguese, and Galician. Given a sentence containing a target MWE, the task is to determine whether the expression is used idiomatically or literally. The task is inherently context-dependent: the same MWE may have different meanings depending on its usage. Therefore, models must rely on contextual information rather than treating expressions as fixed lexical units. Two experimental settings are considered to assess generalisation: zero-shot and one-shot. We evaluate model performance using the macro-averaged F1 score to account for imbalanced data.

4 Data

The dataset used in this project is provided by the SemEval-2022 shared task on Multilingual Idiomaticity Detection (Madabushi et al., 2022)[1]. Each instance consists of a target multiword expression (MWE), the sentence in which it appears ("Target"), and its surrounding context ("Previous" and "Next" sentences). Each sample is annotated with a binary label, where 0 indicates idiomatic usage and 1 indicates literal usage. The dataset covers three languages: English (EN), Portuguese (PT), and Galician (GL). The distribution is highly imbalanced, with English accounting for approximately 71% of the samples, Portuguese for 28%, and Galician for only 1%. The task provides two experimental settings, zero-shot and one-shot. In the zero-shot setting, MWEs in the training set do not appear in the development or test sets. In the one-shot setting, the training data includes one idiomatic and one literal example for each MWE appearing in the evaluation sets, allowing the model to observe minimal supervision per expression.

[1] https://github.com/H-TayyarMadabushi/SemEval_2022_Task2-idiomaticity/tree/main/SubTaskA

Data sets	MWEs	Languages			All
		English	Portuguese	Galician	
Zero-shot train	386	3160	1215	51	4426
Zero-shot dev	50	369	141	6	516
Zero-shot test	50	424	207	6	637
One-shot train	434	3200	1217	51	4468
One-shot dev	82	363	141	6	510
One-shot test	70	390	205	6	601

Table 1: Dataset statistics for zero-shot and one-shot sets across MWEs and languages.

4.1 Data Preprocessing and Splitting

The SemEval shared task provides one zero-shot training set (4,491 samples), two one-shot training sets (209 samples including Galician and 140 samples excluding Galician), and a development set containing 739 samples.

We first analysed the datasets and performed preprocessing steps including replacing missing values with (" "), Unicode normalisation (NFKC), removal of control and formatting characters, standardisation of quotation marks, and whitespace normalisation.

As the shared task does not provide labelled test data, all available datasets were merged into a unified dataset without duplicates. The resulting dataset contains 273, 163, and 50 unique MWEs for English, Portuguese, and Galician, respectively. Following the shared task design, new development and test splits were constructed by selecting 50 MWEs for each, distributed proportionally across languages, resulting in 28 MWEs for EN, 17 for PT, and 5 for GL in each split. For the zero-shot setting, MWEs assigned to the development and test sets were excluded from the training set. For the one-shot setting, training data was constructed by moving one idiomatic and one literal instance per MWE from the evaluation sets into the training set; when only one label was available, a single instance was used.

To maintain comparable distributions between zero-shot and one-shot settings, we rebalanced samples across splits by matching language, label, and instance counts where possible. The overall distribution between the zero-shot and one-shot settings remains highly similar, as shown in Table 1.

4.2 Data Statement

Following Bender and Friedman (2018), we note: the corpus is English-dominated and Galician-sparse (<1%), posing training and evaluation challenges for low-resource languages. EN and PT MWEs are from the Noun Compound Senses dataset with human compositionality ratings; GL MWEs are from Wikipedia and CC-100, curated by a domain expert (Madabushi et al., 2022), creating cross-lingual sourcing asymmetry. Annotations from two annotators (disagreements resolved by discussion) lack demographic documentation, introducing potential bias. Binary labelling collapses a continuous compositionality spectrum (Fazly et al., 2009). Galician data scarcity limits generalisability and cross-lingual reliability.

5 Experiments

5.1 Models

We compare three model families spanning non-contextual surface baselines and fine-tuned multilingual transformers. This design allows us to isolate the effects of contextual representations independently of task-specific fine-tuning.

Logistic Regression We use logistic regression as a simple non-contextual baseline with two input representations: (i) TF-IDF features and (ii) sentence representations obtained by combining Word2Vec embeddings with TF-IDF weights. This baseline is designed to be interpretable, reproducible, and consistent across languages. To retain full control over the setup, we implemented the model ourselves and verified its behaviour against scikit-learn. In the TF-IDF setting, texts are represented as weighted bag-of-words features with tunable n-gram ranges. In the Word2Vec variant, word embeddings are trained on the same corpus and aggregated into sentence vectors using TF-IDF weighting. Both variants use a Unicode word-boundary tokenizer with lowercasing; punctuation and single-character tokens are discarded. Optimisation is performed using gradient descent with L2 regularisation. Class weights are not introduced to maintain comparability with contextual models. The surface form of the target MWE is intentionally retained to assess the extent to which lexical patterns alone drive predictions.

mBERT We fine-tune the `bert-base-multilingual-cased`[2], a 12-layer, 768-dimensional encoder pre-trained on 104 languages using masked language modelling and next-sentence prediction. We use the cased variant throughout, as capitalisation provides important cues for idiomaticity (e.g., distinguishing *Double*

[2] https://huggingface.co/google-bert/bert-base-multilingual-cased

Dutch as a named entity from *double dutch* as an idiom).

ModernBERT We additionally fine-tune `modernBERT-base`[3], a 22-layer encoder with 149M parameters, pre-trained on a large corpus of English text and code. ModernBERT incorporates architectural improvements such as rotary positional embeddings and Flash Attention, and supports 8192 tokens.

5.2 Input Representation

Each experiment is defined by an input-variant configuration specifying how dataset fields are assembled into the model input.

Context: we evaluate two settings, `target`, which includes only the sentence containing the MWE, and `previous_target_next`, which includes the target sentence together with its preceding and following sentences (full context), concatenated using `[SEP]`.

MWE segment: the raw MWE string is prepended as a separate segment (`MWE [SEP] context`), providing the model with an explicit representation of the expression prior to contextual encoding.

Transform: the MWE span within the target sentence is either wrapped with special tokens `<MWE>` and `</MWE>` (`highlight`) or replaced with `[MASK]` (`mask`) to remove lexical cues. The special tokens are added to the tokenizer vocabulary as `additional_special_tokens` to ensure they are treated as atomic units.

5.3 Feature Augmentations

We incorporate linguistic features as optional input augmentations.

Named Entity Recognition (NER): NER is applied using language-specific models. For English and Portuguese, we use `wikineural-multilingual-ner`[4] (Tedeschi and Navigli, 2022). As this model is not trained on Galician and exhibited a high false positive rate in preliminary experiments, we instead use a dedicated Galician model, `bert-base-gl-SLI-NER`[5].

[3] https://huggingface.co/answerdotai/ModernBERT-base

[4] https://huggingface.co/Babelscape/wikineural-multilingual-ner

[5] https://huggingface.co/marcosgg/bert-base-gl-SLI-NER

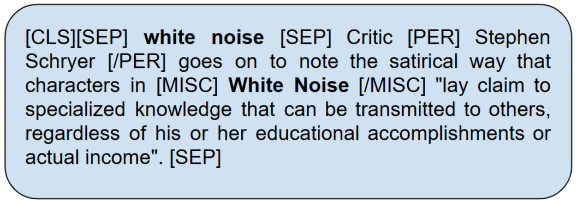

Figure 1: Example of named entity tags appended inline to the input.

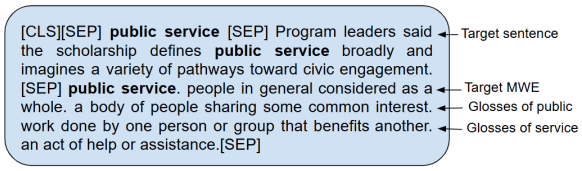

Figure 2: Example of gloss augmentation appended to the input.

Both models follow the BIO tagging scheme. Detected entities are annotated inline using typed tags (e.g., `[LOC]`, `[PER]`), allowing the model to distinguish idiomatic expressions from named entity usages.

Gloss augmentation: Glosses for each word in the MWE are extracted from Open Multilingual WordNet via NLTK (Hauer et al., 2022). As the primary sense is not always listed first, we include the first two glosses per word. These are appended after the context and MWE, as illustrated in Figure 2.

5.4 Evaluation Slices

To separate contextual disambiguation from surface-level memorisation, we construct diagnostic evaluation slices. To construct ambiguous slices, we identify language–MWE pairs that occur with both labels and retain only those appearing at least five times, ensuring recurring ambiguity rather than incidental variation. All corresponding instances form the ambiguous slice ($n = 359$), which is further divided into *majority* ($n = 278$) and *minority* ($n = 81$) instances based on label frequency. Minority-label instances are particularly diagnostic, as models relying on frequency priors systematically fail on these cases and must instead rely on context to override the dominant sense. We construct a matched control slice ($n = 235$) from non-ambiguous MWEs, sampling to preserve language distribution and MWE frequency bins. This ensures that differences reflect ambiguity rather than confounding factors such as imbalance or frequency. Taken together, performance across the

control, ambiguous, and minority slices provides a measure of model behaviour, while high control performance can be achieved through surface heuristics, success on ambiguous and minority instances requires genuine contextual disambiguation.

5.5 Experimental Conditions

All experiments follow a unified pipeline in which each run is defined by a configuration specifying the model, input variant, language scope, and supervision setting. For each configuration, the corresponding data split is loaded, the input representation is constructed (Sections 5.2–5.3), and the model is trained and evaluated under identical conditions.

Hyperparameters are tuned only for the base configuration (full context, no features, no transform) of each monolingual and multilingual setting using development macro-F1. All other configurations reuse these settings to isolate the effect of input variations without introducing additional optimisation noise. Early stopping is applied based on development macro-F1. The hyperparameter tuning grids for all the models are in Appendix A All main runs use random seed 51; additional seeds are used for Galician cross-lingual experiments due to the small test set.

We evaluate three language modes: *monolingual*, where models are trained and evaluated on a single language (EN or PT); *multilingual*, where models are trained jointly on EN, PT, GL and evaluated on both the joint test set and per-language subsets; and *cross-lingual*, where models are trained on source languages and evaluated on a held-out target language with unseen MWE types.

For cross-lingual evaluation, we consider two settings: (i) EN $\rightarrow$ PT (test $n = 207$, seeds = 51), trained on English and evaluated on Portuguese test samples; (ii) EN+PT $\rightarrow$ GL ($n = 63$, seeds 7, 42, 51, 528, 1289), evaluated on all available Galician instances to obtain a stable estimate, as the original test set for Galician only contained 6 test samples, which produced high variance.

6 Results

We evaluate all experimental configurations using macro-F1 as the primary metric, which assigns equal weight to both classes and is appropriate for imbalanced data and for the emphasis on minority-sense instances. Our analysis is organised around the five research dimensions introduced in Section 1: surface-level feature effectiveness, contextual disambiguation, input representation, one-shot adaptation, and cross-lingual transfer.

6.1 Can Surface-Level Features Detect Idiomaticity?

In the zero-shot monolingual setting without feature augmentation, Logistic Regression TF-IDF achieves a macro-F1 of 0.6743 on English and 0.3802 on Portuguese; Logistic Regression Word2Vec achieves 0.6250 and 0.3802 respectively (Table 2). The near-identical performance of the two variants indicates that surface-level lexical statistics capture most of the available signal, with distributional embeddings providing negligible additional benefit at this scale.

For English, logistic regression achieves moderate performance (0.62-0.67 macro-F1), suggesting that idiomatic expressions exhibit recurring lexical co-occurrence patterns detectable by n-gram features. However, inspection of the confusion matrix reveals a systematic bias toward the idiomatic class of TF-IDF, indicating reliance on surface co-occurrence patterns rather than genuine contextual disambiguation. The model correctly identifies frequent idiomatic forms but accumulates errors on literal usages that require overriding this surface prior.

In contrast, both baseline variants collapse on Portuguese, producing identical macro-F1 scores of 0.3802. Both models classify every Portuguese test instance as literal (TP=0, FN=80), resulting in zero recall for the idiomatic class. This degenerate behaviour arises from two compounding factors: the zero-shot constraint prevents any lexical overlap between training and test MWEs, and Portuguese morphological inflection fragments n-gram features across word forms (e.g., *dar uma mão* 'give a hand' versus *deu uma mão* in past tense). Consequently, the Portuguese macro-F1 of 0.3802 reflects the recall of the majority literal class only, not meaningful classification. The magnitude of the EN-PT gap demonstrates that surface features are effective only when lexical patterns are stable across training and test distributions. Figure 3 summarises model performance across languages and settings.

Model	Language	Setting	Macro-F1	Precision	Recall	Accuracy	Confusion Matrix			
							TP	FP	TN	FN
Logistic Regression TF-IDF	EN	zero-shot	0.6743	0.6751	0.6759	0.6745	137	78	149	60
Logistic Regression W2V	EN	zero-shot	0.6250	0.6279	0.6280	0.6250	132	94	133	65
mBERT	EN	zero-shot	**0.8960**	0.8958	0.8977	0.8962	181	28	199	16
ModernBERT	EN	zero-shot	0.8505	0.8509	0.8502	0.8514	164	30	197	33
Logistic Regression TF-IDF	PT	zero-shot	0.3802	0.3068	0.5000	0.6135	0	0	127	80
Logistic Regression W2V	PT	zero-shot	0.3802	0.3068	0.5000	0.6135	0	0	127	80
mBERT	PT	zero-shot	0.6050	0.6045	0.6080	0.6184	45	44	83	35
mBERT	EN	one-shot	**0.9511**	0.9505	0.9528	0.9513	175	14	196	5
mBERT	PT	one-shot	0.7222	0.7275	0.7189	0.7415	49	23	103	30

Table 2: Results for all models across languages and settings. Rows are grouped by language (EN, PT) and setting (zero-shot, one-shot). TP/FP/TN/FN denote true/false positives/negatives.

Figure 3: Comparison of model performance (Macro F1) across languages under zero-shot and one-shot settings using the base configuration (full context, no transformation, no additional features).

6.2 Do Contextual Models Enable Contextual Disambiguation?

Fine-tuned mBERT substantially outperforms both logistic regression baselines in the base configuration (full context, no transform, no feature augmentation). In the zero-shot setting, mBERT achieves macro-F1 of 0.8960 on English, an improvement of +0.222 macro-F1 points over the TF-IDF baseline. Unlike the baselines, mBERT produces a balanced confusion matrix, correctly classifying both idiomatic and literal instances with comparable error rates. This shift from majority-class prediction to balanced disambiguation is illustrated in Table 2. It is consistent with mBERT's pretraining objective: multilingual masked language modelling across 104 languages exposes the model to idiomatic expressions across diverse contexts, enabling its representations to distinguish meaning-in-context from surface form (Devlin et al., 2019; Madabushi et al., 2021). Despite these gains, a substantial EN-PT gap remains: mBERT achieves macro-F1 0.6050 on Portuguese under the same configuration. This gap reflects both data quantity and representation asymmetry: Portuguese accounts for approximately 28% of the training data and receives substantially weaker coverage in mBERT's multilingual pretraining, resulting in less discriminative contextual representations for Portuguese idiomatic expressions.

ModernBERT achieves macro-F1 0.8505 on English in the monolingual zero-shot setting (precision 0.8509, recall 0.8502; TP=164, FP=30, TN=197, FN=33), underperforming mBERT by 0.046 macro-F1 points despite incorporating architectural improvements including rotary positional embeddings, Flash Attention, and extended context length. This result suggests that architectural advances alone do not guarantee improved idiomaticity modelling. A likely explanation is that the SemEval dataset is too small to leverage ModernBERT's architectural advantages, and that mBERT's broader multilingual pretraining provides richer idiom-related representations than ModernBERT's primarily English-and-code corpus. The higher false-negative count (33 vs. 16) indicates that ModernBERT is more conservative in predicting the idiomatic class, consistent with a stronger prior toward the literal majority sense.

In the multilingual joint training condition (EN+PT+GL), mBERT achieves an overall macro-F1 of 0.823, with per-language scores of EN=0.896, PT=0.605, and GL=0.486. English performance is stable under joint training, while Galician reaches near-chance performance due to its extreme data sparsity (fewer than 1% of all instances). The complete results are provided in Table 2.

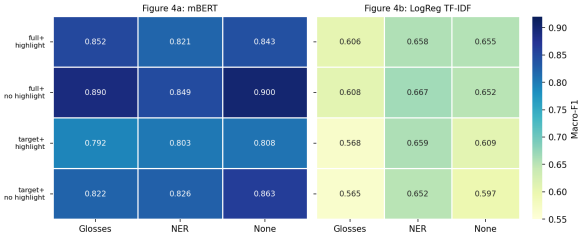

Figure 4: Performance heatmaps (macro-F1) showing the effect of context scope (full vs. target-only), MWE highlighting, and feature augmentation (glosses, NER, none) for mBERT and Logistic Regression TF-IDF.

6.3 How Do Input Representations Affect Performance?

We analyse the impact of context scope, MWE highlighting, and linguistic feature augmentation through controlled ablations in English (EN) and Portuguese (PT). Figure 4 summarises macro-F1 across context $\times$ transform configurations for all feature conditions (none, NER, glosses), for both mBERT and Logistic Regression TF-IDF.

Context scope. Expanding from the target sentence to full three-sentence context (previous_target_next) consistently improves performance. For mBERT (EN, zero-shot), macro-F1 increases from 0.870 to 0.896 (Δ=+0.026). The gain is larger for Logistic Regression TF-IDF (0.596 to 0.674; Δ=+0.078), indicating stronger reliance on extended co-occurrence patterns, whereas mBERT effectively captures sentence-internal cues via attention. A similar trend holds in Portuguese, where mBERT improves from 0.549 to 0.603, suggesting greater dependence on broader discourse when local signals are weaker.

MWE highlighting. Marking MWEs with `<MWE>` tags degrades mBERT performance in the multilingual setting (0.823 to 0.785; Δ=-0.038), indicating that explicit marking biases attention toward lexical form and interferes with contextual disambiguation. In contrast, Logistic Regression TF-IDF shows slight gains in some settings, as `<MWE>` tokens provide a useful discriminative n-gram feature.

NER augmentation. Adding inline NER reduces mBERT performance in English (0.896 to 0.830; Δ=-0.066), likely due to noise from imperfect entity predictions disrupting already strong contextual representations. In lower-resource settings, NER yields modest gains (PT, GL), suggesting that explicit entity signals help when representations are weaker. Logistic Regression TF-IDF benefits slightly overall, as entity labels act as additional surface features without introducing representational noise.

Gloss augmentation. Appending WordNet glosses consistently degrades performance for both Logistic Regression TF-IDF (0.674 to 0.638; Δ=-0.036) and mBERT (0.896 to 0.865; Δ=-0.031). For surface models, glosses increase feature dimensionality without adding signal. For mBERT, attention mitigates some noise, but longer inputs dilute context near the MWE. Overall, gloss augmentation provides no consistent benefit.

6.4 Does One-Shot Supervision Help?

The one-shot setting augments training with one idiomatic and one literal example per evaluation MWE, providing minimal per-expression supervision. A key caveat is that these examples are drawn from the evaluation set, making the one-shot test set strictly smaller than its zero-shot counterpart. The full test set reduces from 637 to 601 instances; within the ambiguous slice, instances drop from 359 to 208 (42%), and minority-sense instances from 81 to 63 (22%). As a result, the one-shot setting contains fewer of the hardest cases, and improvements should be interpreted as an upper bound on learning gain.

Within these constraints, mBERT shows substantial gains. In English, macro-F1 increases from 0.896 to 0.951 (Δ=+0.055) (Table 2), with false negatives dropping from 16 to 5, indicating improved resolution of minority-sense instances through per-expression anchoring. The gain is larger in Portuguese, where macro-F1 improves from 0.605 to 0.722 (Δ=+0.117), suggesting that one-shot supervision compensates for weaker multilingual pretraining by providing explicit sense anchors absent from contextual representations alone.

In contrast, logistic regression shows negligible improvement (Logistic Regression TF-IDF: 0.674 to 0.677). A single additional example per MWE does not provide sufficient frequency signal for n-gram models, which require repeated co-occurrence patterns to stabilise. The near-zero gain supports the interpretation that mBERT's improvement reflects genuine representation-level learning rather than simply increased supervision.

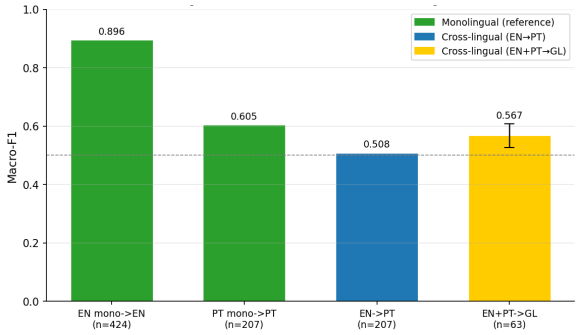

Figure 5: Cross-lingual transfer performance of mBERT under zero-shot base configuration (full context, no transformation, no additional features). Galician results include standard deviation across seeds.

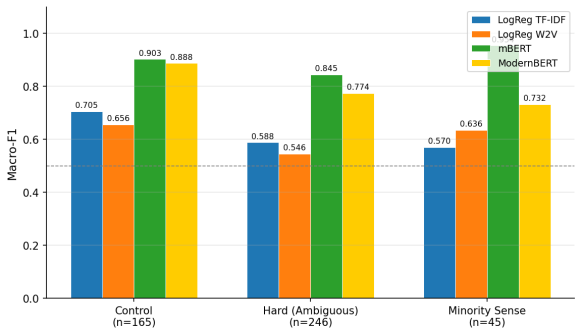

Figure 6: Diagnostic slice performance for EN (zero-shot, base configuration). Macro-F1 across control, hard, and minority MWEs.

6.5 Can Idiomaticity Knowledge Transfer Across Languages?

We evaluate two cross-lingual transfer settings using mBERT under the base configuration. Figure 5 compares transfer performance against monolingual reference scores.

EN → PT. Training on English and evaluating on Portuguese yields macro-F1 0.508 (TP=62, FP=83, TN=44, FN=18), well below the monolingual PT score (0.605) and only marginally above chance. The highly asymmetric confusion matrix - 83 FP vs. 18 FN - indicates systematic over-prediction of the idiomatic class, reflecting transfer of an English-trained idiomaticity prior without appropriate contextual calibration. This failure highlights typological divergence: differences in vocabulary, morphology, and idiomatic constructions limit the transferability of English-trained representations despite mBERT's shared subword space.

EN+PT → GL. Transfer to Galician is evaluated on an extended 63-instance set across five seeds. The standard 6-instance test split yields highly variable results (0.333–0.667) and is insufficient for reliable estimation; all conclusions are based on the extended set (full results in Appendix E). EN+PT → GL achieves mean macro-F1 0.558 (std=0.038), outperforming EN → PT by ∼5 points. This improvement aligns with typological proximity: Galician and Portuguese share vocabulary, morphology, and are mutually intelligible, enabling more effective transfer of idiom-relevant features. In contrast, English-to-Portuguese transfer is constrained by cross-branch lexical divergence. Overall, typological similarity emerges as a stronger predictor of transfer quality than multilingual pretraining coverage alone.

6.6 Diagnostic Slice Analysis: Memorisation or Disambiguation?

We evaluate models on three diagnostic slices from the English zero-shot test set: control (unambiguous MWEs, n=165), hard (MWEs with both idiomatic and literal labels, n=246), and minority (ambiguous instances with the less-frequent sense, n=45). Figure 6 summarises performance.

LogReg TF-IDF degrades monotonically (0.706 → 0.588 → 0.570), indicating strong majority-sense bias: it succeeds on unambiguous MWEs but fails when disambiguation requires overriding the dominant sense. The low minority score confirms systematic majority-class prediction.

mBERT shows a contrasting pattern: performance drops slightly from control (0.896) to hard (0.845), but increases to 0.955 on the minority slice. Higher performance on minority-sense instances than on unambiguous ones provides direct evidence of contextual disambiguation beyond frequency memorisation.

ModernBERT achieves 0.732 on the minority slice, below mBERT (0.955), suggesting weaker idiom-context representations despite competitive overall performance.

Control MWEs are semantically fixed, whereas hard MWEs are genuine polysemes (e.g., *eye candy*, *public service*) requiring context-sensitive interpretation. The collapse of surface models confirms that idiomaticity lies on a compositionality continuum (Fazly et al., 2009).

Frequency-bin analysis. mBERT remains stable across English frequency bins (0.899 for 2–4; 0.896 for 10–19), indicating frequency-robust generalisation, while LogReg TF-IDF is frequency-dependent (0.528 for 2–4; 0.645 for 20–29). Figure 7 shows the English distribution.

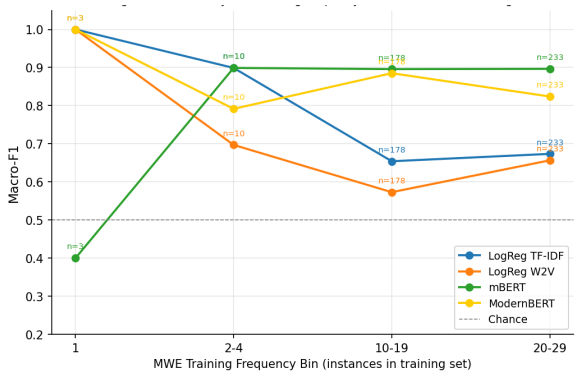

Figure 7: Frequency-bin analysis for English (zero-shot).

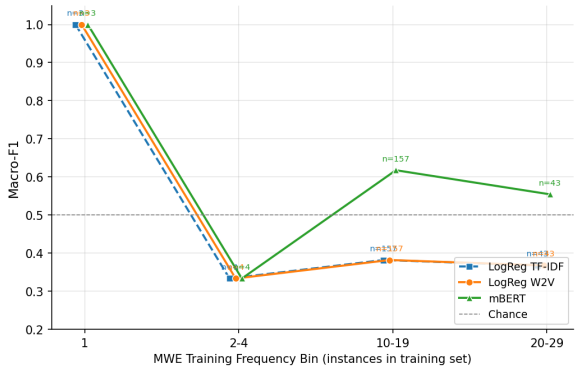

Figure 8: Frequency-bin analysis for Portuguese (zero-shot).

In Portuguese, all models perform well on single-instance MWEs but degrade to 0.33–0.38 in higher-frequency bins. LogReg defaults to majority prediction, while mBERT partially recovers (0.333 → 0.618 → 0.554) but remains below English performance. This reflects morphological variation and data sparsity. Figure 8 shows the Portuguese distribution.

Learning dynamics. mBERT converges rapidly (∼1,500 steps) with steady development gains, indicating representation learning. LogReg shows loss reduction without development improvement, plateauing at ∼0.65 macro-F1. Portuguese mBERT exhibits higher variance and earlier stopping, consistent with weaker training signals (Figure 9).

6.7 Error Analysis: Why and When Do Models Fail?

Error analysis reveals four recurring failure patterns, each tied to a specific linguistic or modelling limitation.

Majority-sense bias in surface models. LogReg models frequently predict the dominant label for minority-sense instances. In Portuguese, this is total: both TF-IDF and Word2Vec classify all 80 minority instances as literal (FN=80, FP=0). In English, the bias is weaker but still evident (FN=60, FP=78). This behaviour is intrinsic to surface models, reflecting reliance on frequency rather than compositional reasoning.

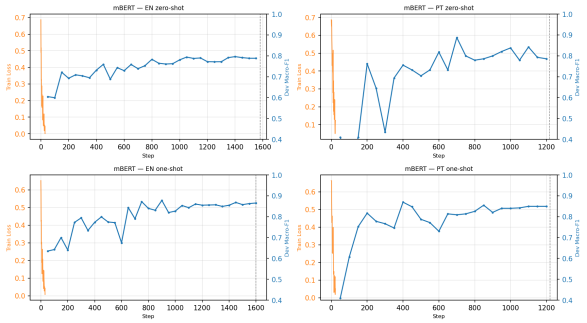

Figure 9: mBERT learning curves for EN/PT.

Named-entity false positives. Both model types misclassify MWEs used as named entities (e.g., *Eye Candy Motorsports*, *Public Service Commission*) as idiomatic due to structural similarity with non-compositional usage. NER augmentation introduces noise for mBERT (macro-F1 drop of 0.066 in English) and yields only marginal gains for LogReg, suggesting that contextual models already encode entity information implicitly. Hard filtering (e.g., Tedeschi and Navigli, 2022) may be more effective.

Cross-lingual sense distribution mismatch. In EN → PT, the asymmetric confusion matrix (83 FP vs. 18 FN) shows that the model transfers an English idiomaticity bias without contextual recalibration. This reflects cross-lingual differences in sense distributions: identical concepts map to different literal/idiomatic proportions across languages, and mBERT remains anchored to the English distribution despite multilingual pretraining.

Subtle contextual failures in mBERT. mBERT occasionally predicts idiomaticity for semantically unusual but compositional phrases, suggesting a conflation of low semantic typicality with non-compositionality. In Portuguese and Galician, additional errors arise from culturally specific expressions whose metaphorical mappings diverge from English, highlighting limits of multilingual pretraining in capturing language-specific figurative meaning.

7 Conclusion

We investigated multilingual idiomaticity detection by comparing surface-level baselines and fine-tuned contextual models across English, Portuguese, and Galician, analysing the effects of input representation, feature augmentation, one-shot supervision, and cross-lingual transfer.

Surface features provide limited, language-dependent signals: logistic regression achieves moderate performance in English (macro-F1 0.674) but collapses to majority-class prediction in Portuguese (0.380), showing that idiomaticity is not reducible to lexical patterns. In contrast, mBERT substantially outperforms surface models and achieves a minority-slice macro-F1 of 0.955 - higher than on unambiguous cases - providing direct evidence of contextual disambiguation beyond frequency memorisation. ModernBERT underperforms mBERT despite architectural improvements, highlighting the importance of multilingual pretraining breadth over model capacity at this scale.

Cross-lingual transfer is asymmetric: EN $\rightarrow$ PT yields near-chance performance (0.508), whereas EN+PT $\rightarrow$ GL achieves 0.558 macro-F1, consistent with typological proximity between Galician and Portuguese. Feature augmentations (NER, glosses) degrade contextual models while offering minor gains for surface baselines, indicating that contextual representations already encode relevant signals but are sensitive to input noise.

One-shot supervision yields substantial gains for mBERT, especially in Portuguese (Δ=+0.117), though results are confounded by reduced test-set size.

Overall, idiomaticity detection is a context-sensitive disambiguation problem. While contextual models perform strongly in English, limitations in cross-lingual transfer, low-resource settings, and ambiguous cases highlight the need for more robust, language-aware representations. Future work should explore typology-aware transfer, parameter-efficient adaptation for low-resource languages, and modelling of the compositionality continuum beyond binary classification.

8 Limitations and Ethical Considerations

Task formulation and linguistic scope. Idiomaticity is inherently a context-dependent and gradient phenomenon rather than a binary property (Fazly et al., 2009). The formulation as a binary classification task necessarily collapses this continuum, obscuring semi-compositional and metaphorical cases. Moreover, idiomaticity is culturally and regionally grounded, and annotations reflect specific language communities; expressions considered idiomatic in one context may be literal in another, introducing systematic bias.

Data and language coverage. The dataset exhibits substantial imbalance across languages, with English dominating and Galician (<1%) severely underrepresented. This imbalance is compounded by asymmetries in multilingual pretraining, where models are exposed to far more English data than Portuguese or Galician. As a result, our findings do not generalise uniformly across languages, and conclusions for low-resource settings should be interpreted with caution. In particular, the extremely small Galician test set limits the reliability of cross-lingual transfer results. Single random seed. All experiments use seed 51; we do not report variance over multiple seeds due to computational constraints. Performance estimates may be noisy,

Evaluation constraints. Our experiments are conducted with a single random seed, and variance across runs is not measured due to computational constraints. This may introduce instability in reported results, particularly for small evaluation subsets such as minority-sense instances and Galician data. Additionally, the one-shot setting introduces a confound: the evaluation set is reduced relative to zero-shot, making it difficult to disentangle improvements due to learning from those due to reduced test difficulty. Finally, the use of macro-F1, while appropriate for class-balanced evaluation, does not capture task-specific costs of different error types.

Generalisability. The dataset is derived from news and web corpora and reflects specific registers of language use. Idiomatic expressions vary across domains, including social media, spoken language, and specialised discourse, and our findings may not generalise to these settings.

Ethical considerations. The dataset may reflect biases present in its source corpora and annotation process, including potential cultural or linguistic biases in what is labelled as idiomatic. Systems trained for idiomaticity detection could also be misused in content moderation or filtering pipelines, where figurative language might be incorrectly flagged or suppressed.

From an environmental perspective, our study involves extensive fine-tuning of large multilingual models across numerous experimental configurations, contributing to computational cost and energy usage. Future work should explore parameter-efficient methods to reduce this footprint.

References

Emily M. Bender and Batya Friedman. 2018. Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics*, 6:587–604.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.

Afsaneh Fazly, Paul Cook, and Suzanne Stevenson. 2009. Unsupervised type and token identification of idiomatic expressions. *Computational Linguistics*, 35(1):61–103.

Bradley Hauer, Seeratpal Jaura, Talgat Omarov, and Grzegorz Kondrak. 2022. UAlberta at SemEval-2022 task 2: Leveraging glosses and translations for multilingual idiomaticity detection. In *Proceedings of the 16th International Workshop on Semantic Evaluation (SemEval-2022)*, pages 145–150, Seattle, United States. Association for Computational Linguistics.

Jisu Kim, Youngwoo Shin, Uiji Hwang, Jihun Choi, Richeng Xuan, and Taeuk Kim. 2025. Memorization or reasoning? exploring the idiom understanding of LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 21678–21699, Suzhou, China. Association for Computational Linguistics.

Harish Tayyar Madabushi, Edward Gow-Smith, Marcos Garcia, Carolina Scarton, Marco Idiart, and Aline Villavicencio. 2022. Semeval-2022 task 2: Multilingual idiomaticity detection and sentence embedding. In *Proceedings of the 16th International Workshop on Semantic Evaluation (SemEval-2022)*, pages 107–121, Seattle, United States. Association for Computational Linguistics.

Harish Tayyar Madabushi, Edward Gow-Smith, Carolina Scarton, and Aline Villavicencio. 2021. AStitchInLanguageModels: Dataset and methods for the exploration of idiomaticity in pre-trained language models. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 3464–3477, Punta Cana, Dominican Republic. Association for Computational Linguistics.

Simone Tedeschi and Roberto Navigli. 2022. NER4ID at SemEval-2022 task 2: Named entity recognition for idiomaticity detection. In *Proceedings of the 16th International Workshop on Semantic Evaluation (SemEval-2022)*, pages 204–210, Seattle, United States. Association for Computational Linguistics.

A Hyperparameter Configurations

A.1 Logistic Regression

TF-IDF features. Table 3 reports the hyperparameter search grid for the TF-IDF variant. The selected configuration corresponds to the multilingual zero-shot base run: unigrams and bigrams (vocabulary size 50,000, `min_df`$= 2$, `max_df`$= 0.90$), 5,000 iterations, learning rate 0.1, no L2 regularisation.

Hyperparameter	Description	Values tried
ngrams	n-gram range	(1,1), (1,2), (2,2)
min_df	Min. document frequency	2, 3, 5
max_df	Max. document frequency	0.85, 0.90, 0.95
norm	Normalisation	l2
smooth_idf	Smoothed IDF	True, False
sublinear_tf	$1 + \log(\text{tf})$	True, False
max_features	Vocabulary size	20k, 30k, 50k
learning_rate	Learning rate	0.05, 0.1
num_iterations	Iterations	3000, 5000
lambda_reg	L2 regularisation	0.0, 1e-5, 1e-4

Table 3: Hyperparameter grid for logistic regression with TF-IDF features.

Word2Vec features. The Word2Vec variant trains embeddings on the preprocessed corpus and constructs sentence representations by averaging word vectors weighted by TF-IDF scores. Table 4 reports the full grid. Selected values: 300-dimensional embeddings, window size 5, 5 negative samples, 30 epochs, sublinear TF scaling.

Table 5 reports the selected configuration for each model and training setting.

Hyperparameter	Description	Values tried
vector_size	Embedding dimensionality	200, 300
window	Context window	3, 5, 7
negative	Negative samples	5, 10
min_count	Min. token count	1, 2, 3
epochs	Training epochs	20, 30, 40
min_df	TF-IDF min. doc. freq.	2, 3, 5
max_df	TF-IDF max. doc. freq.	0.85, 0.90, 0.95
smooth_idf	Smoothed IDF	False, True
sublinear_tf	Sublinear TF scaling	False, True
max_features	TF-IDF vocabulary size	30k, 50k
learning_rate	Learning rate	0.005, 0.01, 0.05
num_iterations	Iterations	3000, 5000
lambda_reg	L2 regularisation	0.0, 1e-5, 1e-4

Table 4: Hyperparameter grid for logistic regression with TF-IDF-weighted Word2Vec features.

B Full Ablation Results

Tables 6 and 7 report macro-F1 for all 12 input-variant configurations of mBERT under the zero-shot monolingual setting for English and Portuguese respectively. Tables 8 provides the corresponding results for Logistic Regression TF-IDF on English. All experiments use seed 51.

C Diagnostic Slice Analysis

Table 9 reports macro-F1 on the three diagnostic slices (control, hard/ambiguous, minority) for all models under the zero-shot base configuration. Slice membership is derived from the zero-shot test analysis: the *control* slice contains unambiguous MWEs ($n = 178$ for EN); the *hard* slice contains ambiguous MWEs- those appearing with both labels at least five times ($n = 246$ for EN, out of 359 total across all languages); and the *minority* slice contains the less-frequent-label instances within each ambiguous MWE ($n = 45$ for EN, out of 81 total). The pattern is consistent across all surface models: macro-F1 drops monotonically from control to hard to minority, confirming reliance on frequency priors. mBERT reverses this trend, achieving its *highest* score on minority instances (0.955), providing strong evidence of genuine contextual disambiguation. ModernBERT occupies an intermediate position: it improves substantially over surface baselines but falls considerably short of mBERT on the minority slice (0.732 vs. 0.955).

D Frequency-Bin Analysis

Tables 10 and 11 report macro-F1 broken down by the training-set frequency bin of each MWE. Bin 1 contains MWEs appearing exactly once in training; bins 2-4, 10-19, and 20-29 correspond to progressively higher frequency ranges.The English table (Table 10) reveals a key asymmetry: surface models perform well at bin 1 (memorisation of single instances) but stagnate at higher frequencies (0.654-0.674 for TF-IDF), whereas mBERT remains stable across all non-trivial bins (0.896-0.899 for 2-4 through 20-29). The Portuguese table (Table 11) shows that all models achieve perfect performance at bin 1 (3 instances; memorisation)

Model	Setting	LR	Batch	Max Len	Epochs	WD
mBERT	Multilingual (EN+PT+GL)	2×10^{-5}	8	256	4	0.01
	Monolingual (EN)	2×10^{-5}	8	128	4	0.01
	Monolingual (PT)	2×10^{-5}	4	128	4	0.01
	Cross-lingual (EN→PT)	3×10^{-5}	4	128	3	0.01
	Cross-lingual (EN+PT→GL)	3×10^{-5}	8	128	5	0.01
ModernBERT	Monolingual (EN)	2×10^{-5}	8	256	3	0.01

Table 5: Selected transformer hyperparameters per model and training setting. LR = learning rate; WD = weight decay.

Context	Transform	Feature	F1	FP	FN
full	none	—	**0.896**	28	16
full	none	glosses	0.891	28	18
tgt	none	NER	0.886	19	29
tgt	none	—	0.870	32	23
full	highlight	glosses	0.865	31	26
full	highlight	—	0.863	40	18
tgt	none	glosses	0.838	29	39
tgt	highlight	glosses	0.832	44	27
full	none	NER	0.830	25	46
full	highlight	NER	0.830	58	14
tgt	highlight	NER	0.814	52	27
tgt	highlight	—	0.774	68	28

Table 6: mBERT zero-shot ablation, English ($n = 424$). "full" = `previous_target_next`; "tgt" = `target`. Sorted by macro-F1 descending.

Context	Transform	Feature	F1	FP	FN
tgt	none	glosses	**0.651**	23	42
tgt	none	NER	0.642	25	42
full	none	NER	0.634	45	29
full	highlight	—	0.617	57	22
tgt	highlight	glosses	0.607	22	49
full	none	—	0.605	44	35
full	highlight	glosses	0.601	35	42
full	highlight	NER	0.592	10	57
tgt	highlight	NER	0.575	51	35
full	none	glosses	0.569	28	51
tgt	none	—	0.538	36	51
tgt	highlight	—	0.465	32	63

Table 7: mBERT zero-shot ablation, Portuguese ($n = 207$). Sorted by macro-F1 descending.

but collapse at bins 10–19 and 20–29, confirming that Portuguese idioms cannot be reliably classified from surface patterns even when training data is available.

E Cross-Lingual Transfer: Per-Seed Results

Table 12 reports complete results for both cross-lingual settings across all seeds and evaluation sets. Due to the extremely small standard Galician test set ($n = 6$), we evaluate on an expanded set of all available Galician instances ($n = 63$) for the primary analysis. Both sets are reported here for completeness.

The EN→PT result (0.508 macro-F1, 83 FP vs. 18 FN) reveals a strong idiomatic bias in the English-trained model: when evaluated on Portuguese, it over-predicts the idiomatic label, reflecting the different sense distribution of Portuguese idioms relative to English. This contrasts with the EN+PT→GL results, where the error pattern is more balanced across seeds, consistent with the higher lexical overlap between Portuguese and Galician.

Context	Transform	Feature	F1	FP	FN
full	highlight	—	**0.684**	75	59
full	none	NER	0.679	73	63
full	none	—	0.674	78	60
full	highlight	NER	0.669	75	65
tgt	highlight	NER	0.660	87	57
tgt	none	NER	0.658	84	61
full	none	glosses	0.639	100	53
full	highlight	glosses	0.634	101	54
tgt	highlight	—	0.616	99	64
tgt	none	—	0.596	106	65
tgt	highlight	glosses	0.591	112	61
tgt	none	glosses	0.583	116	60

Table 8: Logistic Regression TF-IDF zero-shot ablation, English ($n = 424$). Sorted by macro-F1 descending.

Model	Control ($n = 178$)	Hard ($n = 246$)	Minority ($n = 45$)
TF-IDF LR	0.728	0.588	0.570
W2V LR	0.671	0.546	0.636
mBERT	**0.903**	**0.845**	**0.955**
ModernBERT	0.888	0.774	0.732

Table 9: Macro-F1 across diagnostic slices (EN, zero-shot, base configuration). "Hard" = ambiguous MWEs with both labels; "Minority" = less-frequent-label instances within ambiguous MWEs.

Model	Bin 1 ($n = 3$)	2–4 ($n = 10$)	10–19 ($n = 178$)	20–29 ($n = 233$)
TF-IDF LR	1.000	0.899	0.654	0.674
W2V LR	1.000	0.697	0.573	0.657
mBERT	0.400	0.899	0.896	0.897
ModernBERT	1.000	0.792	0.885	0.824

Table 10: Macro-F1 by MWE training-frequency bin, English zero-shot.

Model	Bin 1 ($n = 3$)	2–4 ($n = 4$)	10–19 ($n = 157$)	20–29 ($n = 43$)
TF-IDF LR	1.000	0.333	0.382	0.368
W2V LR	1.000	0.333	0.382	0.368
mBERT	1.000	0.333	0.618	0.554

Table 11: Macro-F1 by MWE training-frequency bin, Portuguese zero-shot.

Direction	Eval set	Seed	F1	FP	FN
EN→PT	PT ($n = 207$)	51	0.508	83	18
EN+PT→GL	GL-6 ($n = 6$)	7	0.667	1	1
	GL-6 ($n = 6$)	42	0.333	2	2
	GL-6 ($n = 6$)	51	0.667	1	1
	GL-6 ($n = 6$)	528	0.625	0	2
	GL-6 ($n = 6$)	1289	0.486	2	1
	Mean (GL-6)	—	*0.556*	*sd = 0.145*	
EN+PT→GL	GL-63 ($n = 63$)	7	0.568	15	10
	GL-63 ($n = 63$)	42	0.624	7	12
	GL-63 ($n = 63$)	51	0.519	6	16
	GL-63 ($n = 63$)	528	0.528	8	15
	GL-63 ($n = 63$)	1289	0.549	18	9
	Mean (GL-63)	—	*0.558*	*sd = 0.042*	

Table 12: Cross-lingual transfer results per seed (mBERT, zero-shot, base configuration). GL-6 = standard 6-instance Galician test split; GL-63 = all available Galician instances. High variance on GL-6 ($\sigma = 0.145$) motivates use of GL-63 ($\sigma = 0.042$) for primary analysis.